

Malikeh Ehghaghi

LinkedIn: [Link](#) | Twitter: [Link](#) | Google Scholar: [Link](#)

Personal Website: [Link](#)

Toronto, ON, CA

Email: malikehehghaghi@gmail.com

Phone number: +1 (647) 9751375

Research Interests

- AI safety and risk mitigation. LLM alignment. Robustness and red-teaming. Safety evaluation and benchmarking. AI governance. Factuality and hallucination mitigation. Long-context learning. Test-time training. Machine unlearning and memorization. Decentralized and modular training. Efficient LLMs. Emergent behaviors in AI systems.

EDUCATION

Ph.D. in Computer Science

September 2025 - Present

Academic Advisor: Colin Raffel

University of Toronto, Department of Computer Science

MSc in Applied Computing

September 2020 – December 2021

Academic Advisor: Frank Rudzicz

Industry Advisor: Jekaterina Novikova

University of Toronto, Department of Computer Science

BSc in Computer Engineering

September 2015 – March 2020

Academic Advisor: Siamak Mohammadi

University of Tehran, Department of Electrical and Computer Engineering

EXPERIENCE

Vector Institute, Toronto, ON, CA

February 2025 – Present

PhD Student (ML Researcher) - Supervisor: Colin Raffel

- **R3 Lab:**

- Conducting research focused on decentralizing, democratizing, and de-risking large-scale AI.
- Active Projects:
 - FineSearch: Collaborative project with the Vanguard team on an end-to-end retrieval-augmented QA system trained on large-scale synthetic, context-driven data. Focuses on memory-efficient encoder-decoder architectures with parallel long-context encoding and linear attention to reduce hallucinations and improve long-context QA performance.
 - TokSuite: Controlled framework for studying the impact of tokenization on language models by training identical models with different tokenizers. Introduces a real-world perturbation benchmark to analyze how tokenization strategies affect performance and robustness systematically.
 - pRisk-Pressure: New metric for adversarial robustness that models LLM safety as a continuous risk-pressure curve instead of fixed-budget attack success rates. Varies attack refinement steps to capture attack cost and efficiency, enabling finer-grained comparisons across models, training methods, and attack strategies.

Arcee AI, San Francisco, CA, US

January 2024 – February 2025

Applied NLP Research Engineer

- **NLP Research Engineering Team:**

- Engineering & Tooling:
 - Contributed to Arcee's MergeKit, the largest open-source model merging toolkit.
 - Contributed to DistilKit, a toolkit for distilling large teacher models into smaller, efficient ones.
 - Built core components of Arcee's SaaS Platform, an end-to-end pipeline for efficient LLM training and inference.
 - Helped develop Arcee Orchestra, an in-house platform for designing agentic workflows.
 - Built multi-GPU training pipelines:
 - On AWS Trainium for domain adaptation of LLMs.
 - On Together AI Cluster using Megatron-LM for training Mixture-of-Experts (MoE) models.
- Project Leadership:
 - Contributed to building an automated workflow generation and large action planning for personal assistant bots in collaboration with SKTelecom
 - Built an agentic HR assistant bot in partnership with Riviera Partners
 - Built function-calling agent models for banking transactions in partnership with American Express (AmEx)
 - Implemented RAG-based Las Vegas tourism search system in partnership with MongoDB
 - Built KidsSafe LLMs with Angel Kids Inc., safe and personalized models for children aged 5–12

- o Multilingual & Social Impact Initiatives
 - Led the Arcee Globe initiative to build culturally and linguistically specific LLMs
 - Released Arcee Meraj and Meraj Mini, top-performing Arabic LLMs
- o Evangelism & Communication
 - Founded and published Arcee AI Docs, including detailed documentation for Arcee Orchestra and Conductor
 - Co-hosted the Small Language Models (SLMs) Show (with Julien Simon), covering SOTA in GenAI
 - Contributed to white papers, technical blog posts, and international conferences
- o Additional Client Collaborations
 - Collaborated with Thomson Reuters, Samsung Electronics, and Emory Health

Lavita, Palo Alto, CA, US

August 2023 – December 2023

Machine Learning Consultant

• **Machine Learning Research Team:**

- o Contributed to designing and implementing a framework for a general-purpose decentralized medical chatbot in partnership with Harvard Medical School and Dartmouth Center for Precision Health and Artificial Intelligence.

University of Toronto, Toronto, Canada

January 2023 – July 2023

Machine Learning Research Team Lead - Supervisor: Prof. En-Shiun Annie Lee

• **Lee Lab:**

- o Led a team of graduate and undergraduate students working on the medical interpretability of clinical notes
- o Benchmarked deep learning models to detect disease labels from clinical notes collected from electronic health records (EHRs)
- o Implemented model-agnostic and model-specific NLP interpretability techniques (e.g., SHAP, LIME, probing, adversarial techniques, etc.) to explore the most important input features in predicting the diagnosis labels
- o Benchmarked interpretability methods based on different interpretability evaluation techniques (e.g., human-centric evaluation, comprehensiveness, sensitivity, etc.)

Winterlight Labs Inc./Cambridge Cognition Inc., Toronto, Canada

April 2021 – July 2023

Machine Learning Engineer - Team Lead: Jekaterina Novikova

• **Machine Learning Research Team:**

- o Led client and research projects with a focus on monitoring and detection of cognitive impairment (e.g., Alzheimer's disease) and mental disorders (e.g., depression, Schizophrenia, anxiety, etc.) from EEG, ECG, speech, and language data in multilingual settings in partnership with Johnson and Johnson and Genentech Roche Group.
- o Developed speaker verification tools for clinical trials to detect duplicate raters and participants with transferability to non-English languages (e.g., German, Danish, Spanish, and Arabic)
- o Analyzed the robustness of the disease detection models to different sources of noise, and developed tools to assess the interpretability and fairness of disease detection models for mental health disorders and cognitive impairment

KavoshComAsia Company, Tehran, Iran

February 2019 – February 2020

Machine Learning Research Assistant – Supervisor: Siamak Mohammadi

• **Research and Development Division:**

- o Designed and implemented on-device ECG heartbeat classification models for arrhythmia, applying deep convolutional neural networks and sequence-to-sequence models in the TensorFlow framework

York University, Toronto, Canada

Summer 2019

Machine Learning Research Intern – Supervisor: Prof. Aijun An

• **Data Mining Lab:**

- o Developed an unsupervised recommender system tool based on the shopping behavior and lifestyle of a group of people in collaboration with Manifold Data Mining Inc.

University of Tehran, Tehran, Iran

July 2018 – February 2019

Machine Learning Research Assistant – Supervisor: Prof. Mehdi Tale Masouleh

• **Human and Robot Interaction Laboratory:**

- o Implemented a motion planning and indoor localization system for Sanbot Robot, applying machine learning algorithms, including reinforcement learning, PSO, etc., in the TensorFlow framework

PEER-REVIEWED PUBLICATIONS

Altıntaş, Gül Sena*, **Ehghaghi, M.***, Lester, B., Liu, F., Zhao, W., Ciccone, M., Raffel, C., "TokSuite: Measuring the Impact of Tokenizer Choice on Language Model Behavior" - ICML 2026 - Under Review

Goddard, C., Siriwardhana, S., **Ehghaghi, M.**, Meyers, L., Karpukhin, K., Benedict, B., McQuade, M., Solawetz, J., "Arcee's MergeKit: A Toolkit for Merging Large Language Models" - EMNLP 2024 Industry Track - Accepted

Ehghaghi, M., Zhou, P., Rajabi, S., Cheng, W., Kuo, C., Lee, E., "Interpretable Disease Prediction from Clinical Text by Leveraging Pattern Disentanglement", IEEE BHI 2023 - Accepted

Ehghaghi, M., Stanojevic, M., Akram, A., and Novikova, J., "Factors Affecting the Performance of Automated Speaker Verification in Alzheimer's Disease Clinical Trials", 5th Clinical Natural Language Processing Workshop, ACL 2023 - Accepted

Ehghaghi, M., Rudzicz, F., and Novikova, J., "Data-driven Approach to Differentiating between Depression and Dementia from Noisy Speech and Language Data", 8th Workshop on Noisy User-generated Text, COLING 2022 - Accepted

Tasnim, M., **Ehghaghi, M.**, Diep, B., and Novikova, J., "Depac: a corpus for depression and anxiety detection from speech", 8th *Workshop on Computational Linguistics and Clinical Psychology*, NAACL 2022 - Accepted

Ehghaghi, M., Novikova, J., Sett, A., Hejrati, M., Robin, J., Teng, E., and Hashemifar S., "Benchmarking Prognostic Longitudinal Machine Learning Models of Alzheimer's Disease Using Speech Features", 1st Workshop on Applications of Medical AI, MICCAI 2022 - Accepted

PREPRINTS

Akram, A., **Ehghaghi, M.**, Stanojevic, M., and Novikova, J., "Zero-Shot Multilingual Speaker Verification in Clinical Trials" - Under Review (Available on arxiv.org)

Gauthier-Caron, T., Siriwardhana, S., Stein, S., **Ehghaghi, M.**, Goddard, C., McQuade, M., Solawetz, J., Labonne, M., "Merging in a Bottle: Differentiable Adaptive Merging (DAM) and the Path from Averaging to Automation" - Under Review (Available on arxiv.org)

WORKSHOPS

- The First Workshop on Machine Learning for Cognitive and Mental Health (ML4CMH), AAAI 2024 - Program Co-Chair

INVITED TALKS

- "Women in Industry Research", Build With AI: Women in Tech (WIT) conference 2026 by Google Developer Groups (University of Toronto) and ACM-W: ACM's Women in Computing - Panelist
- "Model Merging: Theory, Practice and Applications", In-person Tutorial, NeurIPS 2025 - Speaker
- "Scaling Down, Powering Up: Can Efficient Training Beat Scaling Laws", In-person Workshop, Toronto Machine Learning Summit (TMLS) 2025 - Speaker
- "Scaling Down, Powering Up: Can Efficient Training Beat Scaling Laws", Virtual Event, MLOps World | Gen AI Summit - Speaker
- Panel Discussions with Female Leaders in the GTA in Tech, Girls in Tech Conference (GITCon) 2025 - Panelist
- "From Bias to Brilliance: How Women Are Rewriting the Rules of AI", Women in Tech (WIT) Conference 2025 by Google Developer Groups (GDG) and ACM-W: ACM's Women in Computing - Speaker
- Women in AI Research (WiAIR) Podcast, 2025 - Co-host
- "Low Resource Languages", Social Event, ACL 2023 - Main organizer
- "Interpretability and Fairness of the Large Language Models", Birds-of-a-Feather Session, ACL 2023 - Main organizer
- "Factors Affecting the Performance of Automated Speaker Verification in Alzheimer's Disease Clinical Trials" paper at 5th Clinical Natural Language Processing Workshop, ACL 2023 - Presenter
- "Data-driven Approach to Differentiating between Depression and Dementia from Noisy Speech and Language Data" paper at 8th Workshop on Noisy User-generated Text, COLING 2022 - Presenter
- TUSK Series in association with the University of Toronto & Vector Institute, Fall 2022 - Industry speaker
- "Interpretability of Machine Learning Models" Webinar for Computer Webinar Series (AI & Machine Learning Section) at Amirkabir University of Technology, Summer 2022 - Guest instructor
- Wo(Men) In Tech Program, Google Developer Student Clubs, Fall 2021 - Industry speaker
- The Small Language Model (SLM) Show by Arcee.ai, 2024 - Co-host

TEACHING EXPERIENCE

Agentic AI Bootcamp 1.0 and 2.0, Vector Institute, Toronto, Canada
Technical facilitator

Summer 2025 - Winter 2026

Agentic AI Evaluation Bootcamp 1.0, Vector Institute, Toronto, Canada
Technical facilitator

Winter 2026

University of Toronto in Partnership with Vector Institute, Toronto, Canada
Instructional Assistant

March 2023 – April 2023

- TUSK Upskilling Series: Machine Learning Software Foundations - Winter 2023, Spring 2023

University of Toronto, Toronto, Canada
Graduate Teaching Assistant

- CSC413/2516: Neural Networks and Deep Learning - Fall 2025
- CSC457H1: Principles of Computer Networks - Fall 2020
- CSC343: Introduction to Databases - Spring 2021
- CSC263: Data Structures and Analysis - Fall 2021

University of Tehran, Tehran, Iran
Volunteer Teaching Assistant

Spring 2017 – Fall 2019

- Discrete Mathematics - Spring 2017, Fall 2017, Spring 2018, and Fall 2018
- Operating Systems - Fall 2018, Spring 2019, Fall 2019
- Theory of Languages and Automata - Spring 2018
- Data Structure - Spring 2018 and Fall 2018
- Engineering Probability and Statistics - Fall 2018, and Fall 2019
- Database Systems – Fall 2019
- Electronic Circuits - Spring 2017

TECHNICAL SKILLS

Mastered Languages

- Python

Knowledgeable In

- Java
- R
- C
- C++
- MATLAB
- SQL

Data Science

- Web Scraping
- Data Cleaning
- Data Wrangling
- Data Exploration
- Data Visualization
- Statistical Modeling and Prediction
- Database Architecture
- Data Parsing and Filtering
- Big Data
- Time-Series Data Analysis

Deep Learning

- ANN
- CNN
- RNN
- GRU
- LSTM
- Transformers
- GAN
- Reinforcement Learning
- Q/Q*-Learning
- Multimodal Learning
- Signal Processing

NLP

- Language Modeling
- Large Language Models
- LLM Agents
- Agentic Workflows
- Mixture of Agents
- LLM Reasoning
- LLM Planning
- Retrieval-Augmented Generation (RAG)
- Sequence Modeling and Recurrent Networks
- Conditioned Text Generation
- Knowledge Editing
- Pruning
- Distillation
- Continual Pretraining
- Model Merging
- Alignment and Fine-Tuning
- Transfer Learning
- Multi-task, Multi-domain, and Multilingual Learning
- Prompting / Prompt Engineering
- NLP Interpretability
- Bias and Fairness in NLP
- Text Summarization
- Sentiment Analysis
- Question-Answering
- Text Classification
- Information Retrieval
- Machine Translation
- Model Inference
- Modular ML
- Mixture of Experts

Research

- Presentation Skills
- Project Management
- LaTeX
- Team Work
- Problem-Solving
- Paper Writing Skills

Packages

- PyTorch
- Hugging Face
- Megatron-LM
- Transformers
- LM-Eval-Harness
- Alignment-Handbook
- Axolotl
- TensorFlow
- Keras
- Scikit-Learn
- NumPy
- SciPy
- Pandas
- Beautiful Soup
- Matplotlib
- Seaborn
- NLTK
- SpaCy
- Stanza
- Stanford NLP
- Google Lens
- Tesseract
- LlamaParse
- Unstructured
- LangGraph
- LangChain
- Dify
- vLLM

DevOps

- Git
- AWS Glue
- AWS SageMaker
- AWS Trainium
- PySpark
- Docker
- ClearML
- CircleCI
- Amazon EC2
- Amazon ECR
- Runpod
- Cloud Computing

HONORS and AWARDS

- Fully-funded admission in computer science Ph.D. program, University of Toronto, 2025
- MITACS Accelerate Scholarship, University of Toronto, 2021
- Admission to the MSc in Applied Computing (data science concentration) program, University of Toronto, 2020
- Fully-funded admission in computer science direct Ph.D. program, University of British Columbia, 2020
- Fully-funded admission in computer engineering direct Ph.D. program, University of Waterloo, 2020
- Fully-funded admission in MMath in data science program, University of Waterloo, 2020
- Vector Scholarship in Artificial Intelligence, University of Waterloo, 2020
- Ranked 1st among all students of BSc in computer engineering entered in 2015, University of Tehran, Iran
- Success in math and computer olympiads in Iran for two successive years, 2013 & 2014
- Ranked 1st among 2000 candidates at the entrance examination of Kherad Institute, 2011
- Certificate of distinction at the Australian Mathematics Competition, 2009

CERTIFICATES

- Data or Specimens Only research, CITI Program, Jan 2023
- Mitacs Accelerate Program, Feb 2022
- Advanced Linear Models for Data Science 1, Johns Hopkins University, Coursera Certificate, Aug 2020
- Mathematical Biostatistics Boot Camp 1, Johns Hopkins University, Coursera Certificate, Aug 2020
- Mathematical Biostatistics Boot Camp 2, Johns Hopkins University, Coursera Certificate, Aug 2020

Languages

- Farsi - Native proficiency
- English - Professional working proficiency
- Arabic - Limited working proficiency
- French - Limited working proficiency